Representative layout check: WBEA methane monitoring manuscript

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ABSTRACT

1. Introduction

Continuous monitoring networks are increasingly used to interpret methane concentrations near industrial sources (Xu et al., 2026; Saha et al., 2025). Directional and meteorological methods have a long history in air-quality analysis (Uria-Tellaetxe and Carslaw, 2014; Grange and Carslaw, 2019). Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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2. Study area and data

2.1. Monitoring network

The study uses hourly observations from the Wood Buffalo Environmental Association (WBEA) network in northern Alberta, Canada (Fig. 1; Table 1). Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

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2.2. Datasets and temporal design

Datasets and temporal partitions are summarised in Table 2; observation availability is shown in Fig. 2. Sed commodo posuere pede. Mauris ut est. Ut quis

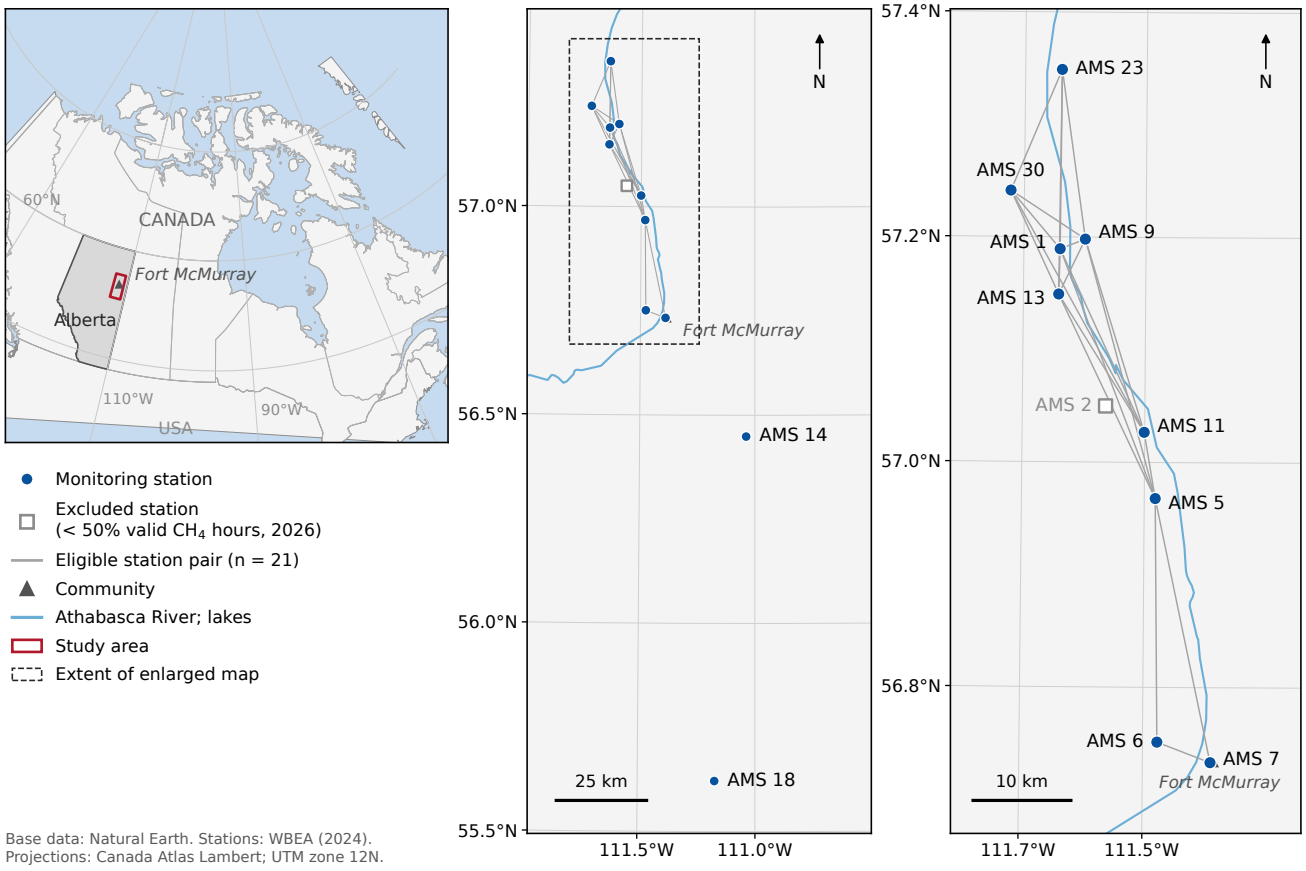


Figure 1: Geographic location and spatial configuration of the Wood Buffalo Environmental Association (WBEA) atmospheric methane monitoring network in northern Alberta, Canada. The maps show the regional location of the study area, the distribution of the 12 monitoring stations, and an enlarged view of the station cluster along the Athabasca River (dashed box). Lines connecting monitoring stations indicate the 21 station pairs eligible for statistical comparison in the 2026 confirmation analysis rather than observed methane transport pathways. The station excluded under the prespecified data-completeness criterion (AMS 2) is shown as an open square. Base data: Natural Earth; station coordinates: WBEA site documents (2024).

Table 1
Geographic characteristics and meteorological measurement heights of the WBEA atmospheric monitoring stations included in the study.

Station	Name	Latitude (°N)	Longitude (°W)	Wind height (m)	Temperature height (m)
AMS 1	Fort McKay - Bertha Ganter	57.1894	111.6406	10	2, 10
AMS 2	Mildred Lake	57.0500	111.5641	10	2
AMS 5	Mannix	56.9680	111.4821	20, 45, 75, 90	2, 20, 45, 75, 90
AMS 6	Patricia McInnes	56.7514	111.4767	10	2
AMS 7	Athabasca Valley	56.7334	111.3905	10	2
AMS 9	Barge Landing	57.1982	111.5995	10, 20	2
AMS 11	Lower Camp	57.0268	111.5008	10	2
AMS 13	Fort McKay South	57.1492	111.6423	10	2
AMS 14	Anzac	56.4489	111.0380	20	2
AMS 18	Stony Mountain	55.6215	111.1728	10, 20	2
AMS 23	Fort Hills	57.3489	111.6397	10	2
AMS 30	Ells River	57.2412	111.7221	10	2

Coordinates (WGS84) from WBEA 2024 site documents. Heights (m above ground) list variables with at least 5% valid hourly values in the 2021–2025 exports; where 10 and 20 m are both listed, the 10 m sensor is used when valid. Methane, total hydrocarbons and non-methane hydrocarbons are reported together at every station (ppm, hourly) by WBEA's continuous hydrocarbon analysers, with an automated zero/span check that occupies one hour per day and is flagged in the data (WBEA SOP-ANA-008; WBEA monthly reports). The analyser model is documented here only for AMS 5 (Thermo 55i, installed 2021); WBEA reports analyser replacements at AMS 5 and AMS 23 in September 2024. AMS 5 has a meteorological tower with sensors at 20–90 m; its 90 minus 20 m temperature difference is the stability measure.

Table 2
Observational datasets and temporal design of the methane-monitoring analysis.

Dataset	Variables (units)	Source	Resolution	Years
Methane	CH_4 (ppm)	WBEA continuous monitoring, final validated data	1 h	2021–2026
Surface wind	Speed ($m\,s^{-1}$) and direction (°)	WBEA continuous monitoring	1 h	2021–2026
Air temperature	Temperature (°C) at 2–90 m; AMS 5 tower gradient	WBEA continuous monitoring	1 h	2021–2026
Boundary-layer height	ERA5 boundary-layer height (m)	Copernicus Climate Change Service, ERA5	1 h	2021–2026
Station locations	Latitude, longitude (WGS84)	WBEA site documents (2024)	–	–

Temporal partition	Dates (MST, hour-ending)	Valid CH_4 hours by station	Use
Development	1 Jan 2021 01:00 – 2 Dec 2025 00:00	90.6–94.2%	Exploratory analysis and method development
Reference	1 Jan 2021 01:00 – 31 Dec 2023 23:00	90.1–94.3%	Fixed station medians, thresholds and stability scaling
Confirmation	1 Jan 2026 01:00 – 1 Oct 2026 00:00	81.5–95.1%[a]	Independent confirmation, sealed until methods were frozen

All timestamps are Mountain Standard Time (UTC−7, no daylight saving) and refer to the end of the hourly averaging period, as in WBEA reporting. Valid CH_4 hours: range across stations of the share of hours with a valid value (zero/span, calibration and invalid hours count as missing). Missing-value handling, station and pair eligibility, and comparison windows are described in Section 3. [a]Excluding AMS 2 (41.2%), which fell below the 50% completeness criterion.

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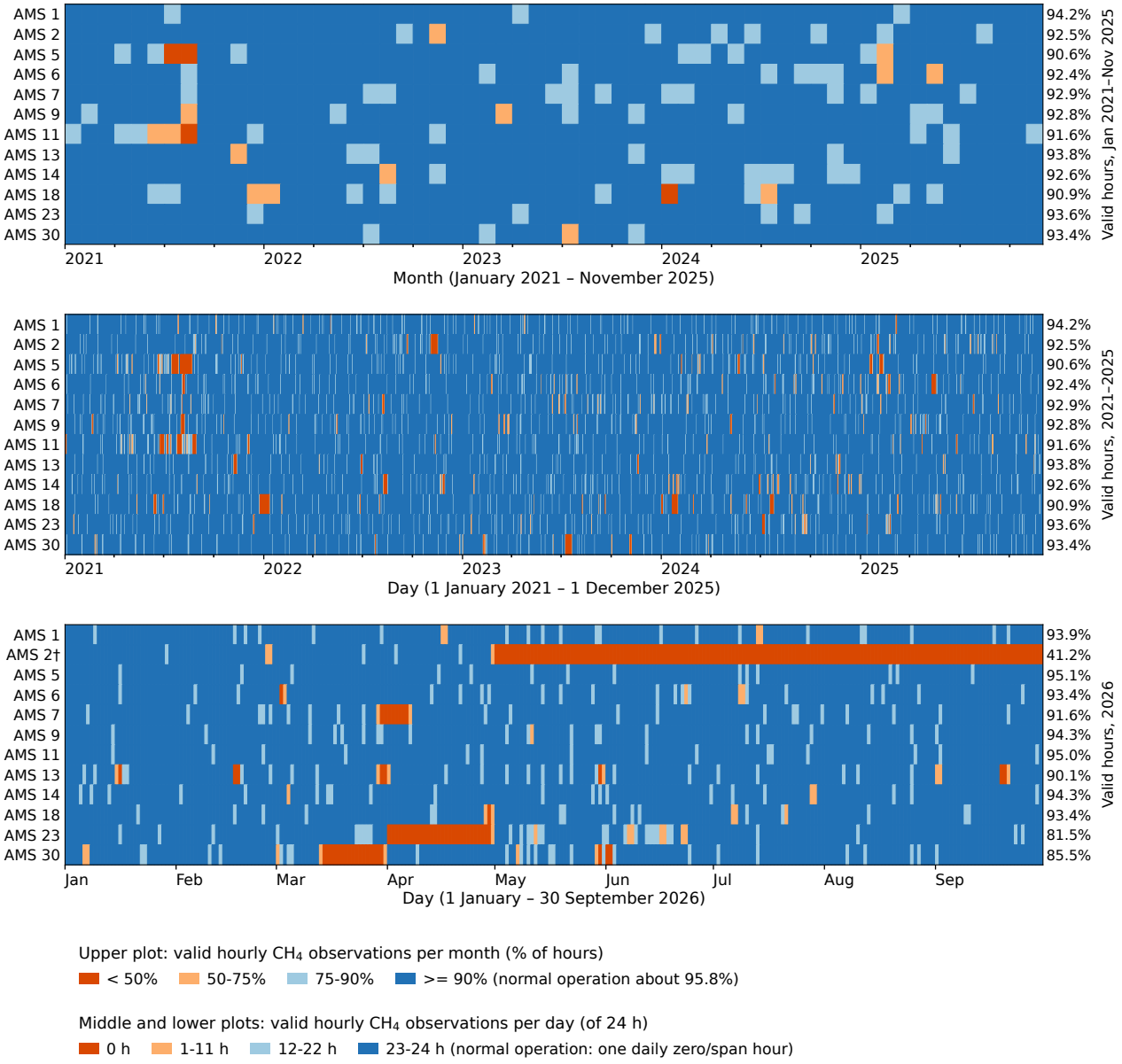


Figure 2: Availability of hourly methane (CH_4) observations at the 12 WBEA monitoring stations. Upper plot: monthly completeness, January 2021 – November 2025 (development period). Middle plot: daily completeness over the same period. Lower plot: daily completeness, 1 January – 30 September 2026 (confirmation period). Completeness is the share of hours with a valid CH_4 value; hours flagged for zero/span checks, calibration, maintenance or invalidity count as missing, and no values are interpolated. Daily cells are classed by the number of valid hours (0, 1–11, 12–22 and 23–24 of 24) and monthly cells by percentage ($<$50, 50–75, 75–90 and $\geq$90%), using the same colours; under normal operation one hour per day is used for the automated zero/span check. Values at right give each station's share of valid hours in the period. Times are Mountain Standard Time, hour-ending. †AMS 2 was excluded from the 2026 confirmation analysis (41.2% valid hours).

3. Methods

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